

Resume Analyzer and Skill Enhancement Recommender System

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Abstract—The research “Resume Parsing and Job Prediction” mainly deals with prediction of an appropriate job title for a person based on their resume. Due to rapid increase in job applicants, companies find it difficult to manually read every resume as this work requires tremendous amounts of time and effort. Hence the resume parser has gained a lot of importance in recent years. Here we use a custom CNN model to predict the job role suitable for an applicant based on their resume. We also compare the model’s performance with other machine learning models Random Forest and SVM and neural network models like custom CNN and a pre trained BERT model. Job applicants also find it difficult to check how compatible they would be for a job. Hence, we use a word2vec vectorization technique and cosine similarity and find the similarity between the resume and a company’s job description. This application also includes a scoring and a feedback system that provides personalized feedback for the applicant’s resume and scores it based on important contents like their education, skills, experience.

Keywords—*resume, CNN, word2vec, cosine similarity, resume scoring, feedback, sentiment analysis.*

I. INTRODUCTION

A resume is said to be more than just a mere paper and is the most important document for an applicant to land a job and for a company to judge the applicant. In recent years and with advancements in Natural Language Processing and Machine Learning, extraction of important information of data from a resume has become widely popular and time saving.

A resume is a document that lacks a proper structure. The headings to be included in a resume are not strict and each resume varies with another in its format, headings, style and even choice of words used and hence resume parsing has always been an area under study.

This research aims to build a comprehensive resume analysis tool and the objectives of this research are:

Given a resume, extract all the text from the document. Filter out unnecessary details and retain necessary ones for job role prediction. Using details like education, skills,

experience, and projects, predict a suitable job role using machine learning models.

Score each resume based on their skills, experience, and education. Check compatibility between a resume of a person and a companies Job Description and provide curated feedback to land the desired job.

Sort all applicants applied for a specific job based on the resume’s score and compatibility for a company to find the most suitable applicant.

The rest of the paper is divided into sections as follows. Section 2 Literature Survey. Section 3 Methodology used. Section 4 Result and Analysis. Sections 5 and 6 include the Future Scope and conclusion of this research.

II. LITERATURE SURVEY

Research on resume parsing and information extraction has yielded promising advancements. Sajid et al. [1] presented an accurate and robust framework for resume parsing in e-recruitment that handles diverse formats and leverages NER for improved candidate profiling. Channabasamma et al. [2] developed a contextual model using SpaCy’s NLP capabilities that captures semantic relationships in resume text for more accurate extraction. Sinha et al. [3] proposed techniques to break down resumes and perform text analysis with Python to reveal insights into candidates. Tikhonova and Gavrishchuk [4] explored NLP methods involving classification and heuristics to automatically segment candidate CVs. Additionally, Gunaseelan et al. [5] implemented a system that effectively extracts segment details from resumes using a novel multi-level classification approach that achieved state-of-the-art accuracy. These works demonstrate the potential for automated resume parsing and analysis to transform recruiting.[14] This paper focuses on utilizing data mining techniques in e-commerce service recommendations. It emphasizes the use of these techniques to enhance user experience by personalizing service offerings based on user preferences and behaviors.[15] explores the application of data mining in analyzing social awareness on platforms like Twitter and Facebook. It employs natural language processing to extract and analyze wearable technology-

related data, highlighting the role of data mining in understanding social trends and awareness. [16] This paper looks into sentiment analysis, a branch of natural language processing (NLP), exploring techniques to automate the process of extracting attitudes from text. It highlights the role of sentiment analysis in social network data, discussing methodologies like AI, dictionary-based methods, and rule-based frameworks for evaluating sentiment in various contexts including social media and political discourse. [18] This paper discusses an algorithm that employs neural networks for language modeling, specifically for predicting the next word in a sequence. The work focuses on the use of Long Short-Term Memory (LSTM) neural network architecture, analyzing its performance on English and French language tasks, and shows significant improvements over standard recurrent neural network models.

Recent research has proposed various techniques for automated resume parsing and screening using machine learning and NLP. Sajid et al. [1] presented a robust framework that handles diverse formats, while Jivtode et al. [6] focused on resume analysis using ML and NLP for recruitment. Goyal et al. [7] extracted resume data by structuring unstructured text. Pudasaini et al. [8] introduced a system for scoring and matching resumes and job descriptions using word2vec and Gale-Shapley algorithm. Furthermore, Sinha et al. [3], [9] proposed methods to break down and screen resumes using NLP and ML, revealing deeper insights. These works showcase the potential of ML and NLP to transform resume analysis and screening.

[19] focus on using Temporal Convolutional Networks (TCN) for identifying offensive content in text. The study benchmarks TCN against other neural network models, including Bidirectional LSTM and GRU, across various word embeddings like GloVe and FastText. It utilizes a dataset of annotated tweets for this purpose, demonstrating TCN's effectiveness and comparable accuracy with other recurrent neural network models in classifying offensive language in digital communication. [20] This paper focuses on performing sentiment analysis on Tamil language online news feeds. It categorizes the characteristic. The study highlights the importance of building an efficient model for constructing a relevant knowledge base in sentiment analysis. [22] Presents a novel approach for product recommendation and rating by analyzing customer reviews from e-commerce websites. The method extracts product specifications and customer reviews, identifying key terms related to the technical features of the product to determine the sentiment of each feature. This sentiment analysis is then used to calculate a score for individual product features, which are aggregated to form an overall product rating.

Automated resume screening powered by NLP and text analytics has transformed traditional recruitment practices. Recently, researchers have developed innovative techniques to parse, extract information from, and evaluate resumes/CVs. Mhaske et al. [10] performed an exploratory analysis of using NLP for resume parsing to improve efficiency. Khan et al. [11] designed a system for resume parsing and summarization leveraging NLP. Taking a different approach, Das et al. [12] built a CV parser model utilizing entity extraction and big data tools for structured information extraction. Moreover, Kudatarkar et al. [13] pioneered an unstructured text analytics methodology to

qualitatively assess resumes based on topics and sentiment. Such advances promise more efficient, accurate and intelligent resume screening at scale.

[23] This paper explores the sentiment analysis of Twitter data related to the demonetization policy implemented in India, using advanced machine learning techniques. The study examines a large dataset of tweets collected during the demonetization period, applying machine learning algorithms to classify these tweets into various sentiment categories like positive, negative, or neutral. Techniques like Natural Language Processing (NLP), along with machine learning models such as Support Vector Machines (SVM) or Neural Networks, are used to analyze and interpret the sentiments expressed in the tweets.

[21] This paper explores the integration of Convolutional Neural Networks (CNN) with K-Means clustering for sentiment analysis. The CNN, a deep learning technique known for its efficiency in handling large datasets and extracting features from text data, is potentially used to process and analyze textual sentiments. K-Means clustering is applied to categorize these sentiments into distinct groups, such as positive, negative, and neutral. The study focuses on the application of this hybrid method to various data sources, like social media posts or customer reviews, offering a more nuanced understanding of public opinion or consumer sentiment.

III. METHODOLOGY

A. DATASET

This research used the following datasets:

1) The “resume dataset” which is a popular opensource dataset available on kaggle <https://www.kaggle.com/datasets/gauravduttakiit/resume-dataset>. This dataset consists of 960 records that show the resume and the job category of a person.

Further, the resumes of about 100 college students from Amrita School of Computing, Bangalore were collected and added to the dataset for better learning for the model.

2) The “Naukri.com: Data Science Job” dataset which is available on kaggle <https://www.kaggle.com/datasets/sanjeevsahu/naukricom-data-science-jobs>. This dataset contains 12000 rows which provide the most important skills required for a particular job position.

3) The “EdX Courses” dataset and the “Coursera Courses” dataset both being available on kaggle <https://www.kaggle.com/datasets/khusheekapoor/edx-courses-dataset-2021>

<https://www.kaggle.com/datasets/khusheekapoor/coursera-courses-dataset-2021>

These datasets contain about 4000 of the most popular courses available on EdX and Coursera with the difficulty of the courses which are useful in the feedback system.

B. Data Preprocessing

Resume is a document that lacks a proper structure, and each resume has its own format and style and hence preprocessing becomes a very important process. The

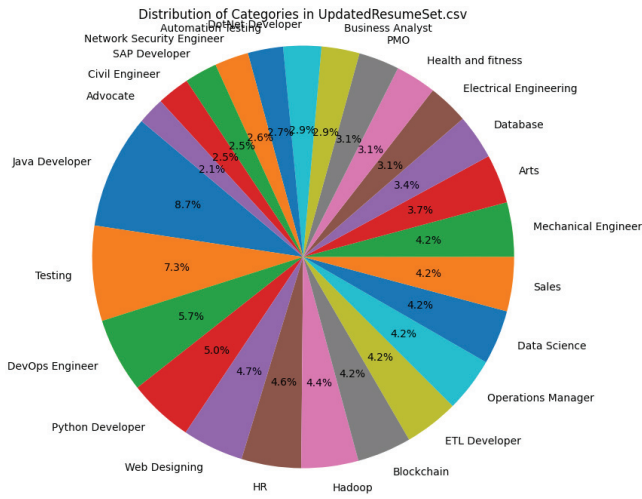


Fig. 1. Job Role Distribution.

content of the resumes was cleaned by removing bulletin points and arrows. Since headings play an important role in extraction of useful content, all headings were obtained, and a colon was added after them. Finally, all the text in the resume was made to lower case which is important for pattern matching.

C. Features extracted

The resume contains all the important details of an applicant, yet all these details are not necessary for job role prediction like their name, sex, phone number. Hence only some of the resume's content was extracted namely content falling under headings like Education, Skills, Work/Experience, Projects/Internships, Certification. The extraction of necessary details and removal of others also helped in reducing computation time and memory.

To obtain the contents under these headings, 2 main approaches were used 1) Regular Expression (Regex) and 2) spaCy.

The Regex worked using basic regular expression technique where each heading was found if the word was followed by a colon while the more advanced spaCy technique was able to find the necessary headings even if those headings were not necessarily followed by a colon :

D. Job role prediction

Prediction of a suitable job role from a resume is an important task as it would give the applicant more insights on which role would be more suitable to them and which jobs, they would be more inclined towards past data. This would help them to prepare better and apply for suitable openings.

To perform this prediction, various models have been studied and compared.

MODELS

1) Random Forest Classifier

The random forest classifier model belongs to the ensemble learning method that works by constructing a decision tree of multiple levels during training to predicts the class that a particular resume would belong to. The random forest model is advantageous to overcome overfitting problem and works well in case of noise and outliers.

The training and testing data were split in ratio of 0.8 to 0.2 and uses a One vs Rest classifier that fits one classifier per class while considering rest of the classes as negative.

The number of decision trees is set to 100 for this experiment.

2) K Nearest Neighbours (KNN)

The KNN model is a machine learning model that predicts the class of a resume by comparing its distance with its neighbours and thus assigning the class of a resume to the class where the distance is less for majority data points. The KNN model is a light weight and easy to understand and easy to run model that can be used when the dataset size is limited.

The training and testing data were split in ratio of 0.8 to 0.2 and the KNN model uses the Euclidean distance for distance calculations.

3) Support Vector Machine (SVM)

The SVM belongs to the machine learning set of models, works by finding the best hyperplane that separates the data into its various classes also by increasing the margins between them. The SVM model is a robust model that aims generalises well to new data and hence reduces overfitting.

The training and testing data were split as 80% to 20% for better training and the SVM uses a linear kernel.

4) Custom CNN

The Convolutional Neural Network (CNN) model belongs to the class of neural network models that is suitable for complex tasks and for training large datasets and natural language processing (NLP) tasks.

The CNN model is effective in capturing words or phrases which are local patterns. This CNN model also works well in finding specific word combinations and identifying proper semantic structure.

The custom CNN model shown in Fig.2 has the following structure. The model has an embedding layer to convert the text to integers. This model also has 1 convolutional 1D layer with 128 filters using a ReLU activation function. This is followed by a max pooling layer which is used to capture the important data and reduces the dimensionality. This model also contains 2 dense layers and a dropout layer with 50% dropout to reduce overfitting.

5) BERT

The Bidirectional Encoder Representations from Transformer (BERT) is a transformer-based model developed in 2018 by Google designed for NLP tasks. It understands the context in a better manner based on its neighbouring words in both directions.

The BERT model is trained on the resume dataset and the training and testing is split into 0.8 to 0.2 ratio for better training.

E. Similarity Calculation

Often, an applicant is unaware if he would be a good fit for the job role provided by the company. So, this similarity calculation technique can be used to find the common features and compatibility between the job role provided and the resume.

This similarity calculation is done using the common cosine similarity. First, both the resume and the job description and converted to vectors using the Word2Vec vectorization technique and then the similarity between the two documents are evaluated. A threshold of 0.5 is kept as a minimum threshold and when the similarity exceeds this threshold, the person is very suitable for the job role.

F. Scoring Of Resume

Scoring of resumes is a very crucial task as it provides a score for each resume based on its content.

The process of scoring is done by providing weights for educational degrees like Bachelors, Masters, PhD and skills and certifications with the weight being higher for more difficult and rare degrees, skills, and certification.

This score can be useful for the applicant as they can see their score and improvise their skills and thus their resume accordingly.

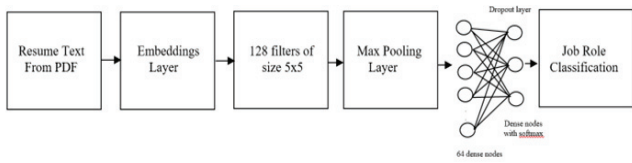


Fig. 2. CNN model architecture

Feedback System

The feedback system provides personalised feedback for an applicant's resume using job description and the content of the resume.

When the similarity between the resume and the job description is lesser than the threshold of 0.5, we recommend courses for those skills not present in the resume but required for the job role. The courses recommended are mainly of beginner level so that the applicant can quickly learn them and increase their score and compatibility.

When the similarity exceeds the threshold of 0.5, the applicant is already in line with the job description. Hence, we recommend courses of intermediate and advanced level, to boost their skillset and to help them land that job by increasing their score.

This feedback system also identifies the tone and the sentiment used in the resume and recommends usage of more positive words if the overall tone of the resume is found to be negative as the use of positive words increases the chance of the resume having more score and importance.

IV. EXPERIMENTATION AND RESULTS

A. Experimentation Setup

In this experiment, the dataset was split in the ratio of 80% to 10% to 10% for training and testing and validation respectively.

The code was processed on a system that has the following configurations Intel i5 8th generation, 8 GB of RAM, Nvidia graphics card for running Cuda and an active internet connection.

B. Hyper Parameters

Table 1 shows the hyper parameters used in the custom CNN model. The Adam optimizer helps to adjust the learning rate for the parameters used and the ReLU activation function is helpful as it adds nonlinearity thus helping it to learn complex patterns in the data.

TABLE I. HYPER PARAMETER USED IN THE CNN MODEL.

EPOCHS	ACTIVATION FUNCTION	OPTIMIZER	BATCH SIZE
30	ReLU	Adam	32

Fig 3 shows the training and validation loss of the custom CNN model.

It is seen from this fig 3 that, the training loss initially is around 3 and higher than the validation loss. But then at around the 20th epoch, the validation loss which was lower crosses the training loss graph. The validation loss trends downwards indicate that the model is generalizing well to new data and unseen data. However, since the validation loss continuous to decrease after the graphs meet, indicate that model might be overfitting to the training data.

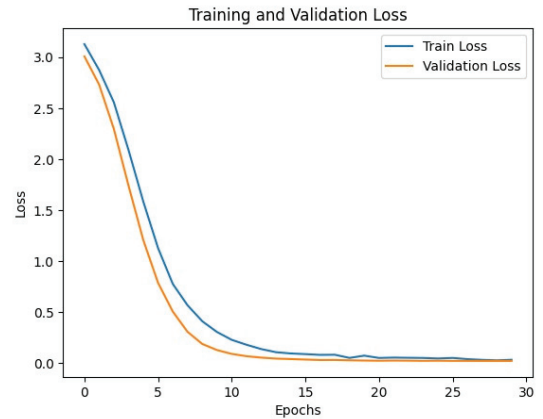


Fig. 3. Loss plot for the CNN model.

Fig 4 shows the accuracy plot for the CNN model for the training and validation sets. The training accuracy increases sharply and quickly reaches an accuracy of 1.0 at around the 20th epoch which shows that the model is learning very quickly. The validation accuracy on the other hand, increases more rapidly and reaches an accuracy of 1.0 and plateaus much earlier at around the 5th epoch.

This indicates that the model might not be learning on new and unseen data and thus after the 20th epoch, the model doesn't learn well. This also suggests that the model might be overfitting the training data as it doesn't generalize as effectively as the validation data.

To reduce the problem of overfitting, regularization methods were employed. The techniques used for this regularization included early stopping methods. Dropout layers with 50% dropout were also added to the CNN model to reduce overfitting by making 50% of the neurons randomly selected and dropped out. This helps the model to have a more robust and general representation as it doesn't rely on a specific neuron.

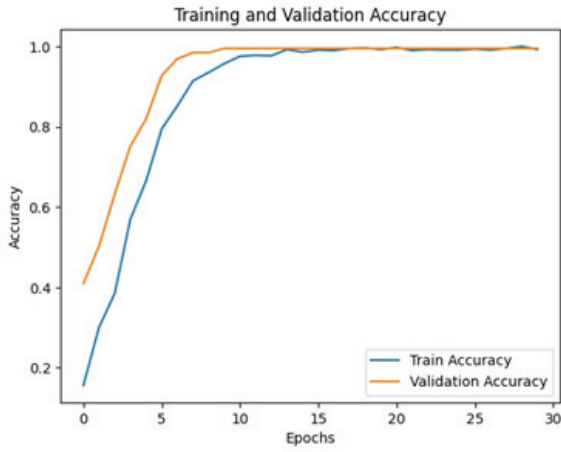


Fig. 4. Accuracy plot for the CNN model

C. Results

Table 2 shows the accuracy and F1 score of the 5 models evaluated. The custom CNN model works the best among the other models as it gives high accuracy and F1 score.

The high accuracy and F1 score shown by the CNN model indicates that the CNN model can perform well even in the case of new and unseen data.

The machine learning models like KNN, SVM and Random Forest also give high and similar accuracy and F1 score values when the dataset size is limited but when the dataset size is increased, the neural network models perform better and give better results.

The CNN and the BERT model show very similar results in terms of accuracy and f1 score but the custom CNN model works faster than the pre trained BERT model and thus gives results with lesser time needed for processing.

TABLE II. ACCURACY AND F1 SCORE OF THE MODELS EVALUATED.

MODEL	ACCURACY	F1 SCORE
CNN	99.48%	0.9950
Random Forest	99%	0.9932
SVM	98.23%	0.9949
KNN	98.44%	0.9839
BERT	99.48%	0.9950

TABLE III. ACCURACY AND F1 SCORE FOR CNN MODELS FOR VARIOUS EPOCH.

EPOCHS	ACCURACY	F1 SCORE
10	96.8%	0.96
20	98.4%	0.982
30	99.48%	0.9950
40	99.48%	0.9948

From table 3, we can see that the CNN model's performance increase through the epochs 10 20 and 30.

In the 10th epoch, the model's accuracy is evaluated as 96.8% which increased to 98.4% in the 20th epoch. The accuracy further increases to 99.48% in the 30th epoch.

The accuracy then remains constant at the 40th epoch which indicates that training the model beyond this epoch may not provide significant change in accuracy.

A similar increase in F1 score is observed through the 10th, 20th and 30th epoch but the F1 score remains constant in the 40th epoch.

This indicates that the 30th epoch is a good number to stop training the CNN model and the model might overfit the data after this epoch.

V. CONCLUSION

This research concludes that the extraction of important details from a resume like education, skills, experience etc help in obtaining a high accuracy. The custom CNN model works the best and gives the highest accuracy of 99.48% and its performance is also comparable with pre trained models like BERT.

These machine learning models work very well and give high accuracy when the size of the dataset is limited. But if the size of the dataset were to be increased to a bigger number, the pre trained model and heavier models like CNN would work better.

The feedback system that provides appropriate courses helps to improve the score of the applicant's resume and the compatibility with the job description thus improving their chances of landing the job.

VI. FUTURE SCOPE

Increase in the size of dataset and training the CNN model on a greater number of resumes would give a better understanding of the exact working of the model and would help to compare it with a pre trained model like BERT.

Resumes lack a proper syntax and format, and the usage of words and headings differ. Using more sophisticated pattern matching techniques could be used to find and extract content under headings with similar meanings when the match isn't exact.

A more sophisticated scoring technique can be employed with more skills and certifications being included so that the resume score would include more scenarios and thus improve the scoring.

The model works accurately and predicts the job precisely when the resume is in English. More languages can be supported so that the model works accurately when the resume is written in another language.

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